

WORKSHOP PLAN

AI Innovation Lab

Learn it. Build it. Launch it.

A fully hands-on AI workshop in which every student team builds, tests and presents its own AI project

Proposed for	Billabong High International School, Jabalpur Satya Prakash Public School, Jabalpur
Conducted by	Baderia Global Institute of Engineering & Management, Jabalpur
Participants	Students of Class 8 to Class 12; up to 40 students per batch, working in teams of 3 or 4
Duration	3 hours in the school computer lab (a 2-hour version is given in Section 10)
BGIEM team	One lead trainer, one co-trainer and three or four student mentors from BGIEM
Lead trainer	Mr. Deepak Vishwakarma, Trainer, BGIEM

1. THE IDEA

Most AI sessions in schools end with students having watched a few impressive demonstrations. This workshop is planned the other way round. Students spend almost the entire time at the computers, working in small teams, and by the end of three hours every team has built and presented an AI project of its own.

Each team works like a young start-up. It picks a real problem from its school or from Jabalpur, studies the problem with AI, builds a working app for it, designs a logo and a poster, plans an automation and finally pitches the whole idea at a Demo Day in front of its teachers. Every AI concept is learnt while doing one of these tasks, at the moment the students need it, rather than in a lecture before it.

By the end of the workshop, every team will have:

1. A machine learning model, trained by the team, that sorts waste and runs on a phone
2. A research notebook on its problem, with a mind map and an AI-made podcast
3. A working web app with a shareable link and a QR code
4. A logo, a poster and a pitch slide for its project
5. An automation plan and an 8-second AI promo video
6. A Demo Day pitch, certificates and a take-home AI Starter Kit

2. WHAT MAKES THIS WORKSHOP DIFFERENT

- **Practical from the first minute.** About 80 percent of the time is spent on the computers, not on slides.
- **Learning by building.** Students learn what training data, bias, prompting and automation mean by actually doing them.
- **Local and real.** The problems come from the students' own school and city, such as waste on campus, the Narmada, traffic at the school gate and the heritage of Jabalpur.
- **Something to show at home.** Every team leaves with working links that open on any phone.
- **Built to continue.** The school receives a lab manual, worksheets and a four-week follow-up plan, so the work does not stop when the workshop ends.
- **Safe by design.** School-managed logins, no personal accounts, and no faces or personal data used in training.

3. WORKSHOP FLOW

The workshop is organised as six missions, followed by a Demo Day. Each mission produces something the team keeps.

Time	Mission	What students do	Tool	Output
0:00–0:10	Opening: Can AI Read Your Doodles?	Play Quick, Draw!, play “Real or AI?”, form teams and pick problem cards	Quick, Draw!	Teams and problems chosen
0:10–0:35	Mission 1: Teach a Machine	Train a model to sort plastic, paper and organic waste; test it for bias; open it on a phone	Teachable Machine	Working waste-sorting model
0:35–0:55	Mission 2: Research Like a Scientist	Study the team’s problem from a source pack; check citations; make a mind map and a podcast	NotebookLM	Notebook, mind map, podcast
0:55–1:10	Mission 3: Prompt Battle	Three rounds of prompt challenges on the big screen	Gemini, Nano Banana	Leaderboard points
1:10–1:20	<i>Break</i>			
1:20–1:55	Mission 4: Build the App	Build, test and improve a web app for the team’s problem; share it through a QR code	Gemini Canvas	Web app with a link
1:55–2:15	Mission 5: Brand It	Create a logo, a poster and a pitch slide; write a prompt for the promo video	Nano Banana, Gemini Canvas	Logo, poster, slide

Time	Mission	What students do	Tool	Output
2:15–2:25	Mission 6: Automate It	Plan an automation; the best plan is built live by the trainer	Google Workspace Studio	Automation plan
2:25–2:50	Demo Day	Two-minute pitch per team, app demo, poster and premiere of the team's AI promo video	Google Flow (videos made by the co-trainer)	Pitch and awards
2:50–3:00	Closing	AI Safety Pledge, certificates, take-home kit and group photograph	–	AI Starter Kit

4. MISSION-WISE LAB PLAN

Each mission is written in the form of a lab practical so that the same sheets can later be used by the school's computer teachers.

Opening: Can AI Read Your Doodles?		10 minutes
Aim	To break the ice and show that AI learns from examples.	
Tools	Quick, Draw! (quickdraw.withgoogle.com), projector	
Procedure	1. Teams play Quick, Draw! for two minutes and try to make the computer guess their drawings. 2. The trainer explains how the game guesses: it has learnt from millions of doodles drawn by people around the world. 3. "Real or AI?": six images are shown on screen and teams vote on each one. 4. Each team picks a problem card (Section 5) and chooses a team name.	
Concept learnt	AI learns patterns from data.	
Wow moment	The computer recognises a half-finished drawing within seconds.	

Mission 1: Teach a Machine		25 minutes
Aim	To train an image recognition model and understand how the training data affects the result.	
Tools	Teachable Machine (teachablemachine.withgoogle.com, no login needed), webcam, waste items brought by the trainers: plastic bottles, paper and fruit peels or leaves	
Procedure	1. Open Teachable Machine and start a new Image Project (Standard image model). 2. Create three classes: Plastic, Paper and Organic. 3. Hold each item in front of the webcam and record 50 to 100 images per class from different angles. 4. Click <i>Train Model</i> and test the model live with the items. 5. Bias test: show the model an item it has not seen before, such as a bottle of another colour or a crumpled sheet of paper. Note where it fails, add more varied images and train it again. 6. Click <i>Export Model</i> and then <i>Upload my model</i>. Turn the shareable link into a QR code and open the model on a phone.	
Output	A working waste-sorting model with a shareable link.	
Concept learnt	Training data, classes, accuracy, bias, and why varied data matters.	
Wow moment	A model the team trained ten minutes ago recognises waste on a phone.	

Mission 2: Research Like a Scientist		20 minutes
Aim	To study a problem from trusted sources using AI, and to check every answer through its citation.	
Tools	NotebookLM; a source pack for each problem (three or four articles or PDFs, prepared in advance by the trainers)	
Procedure	1. Create a new notebook and add the team's source pack. 2. Start an Audio Overview in English or Hindi first, as it takes a few minutes to generate. 3. While it is being made, ask three questions about the problem and open the citation for each answer. 4. Generate a mind map of the topic. 5. Listen to the first minute of the podcast and write the three most important facts on the team worksheet.	
Output	A research notebook, a mind map and an AI podcast on the team's problem.	
Concept learnt	Source-grounded AI, citations and fact-checking.	
Wow moment	Two AI voices discussing the team's own topic as a podcast.	

Mission 3: Prompt Battle		15 minutes
Aim	To learn how to write good prompts through a friendly competition.	
Tools	Gemini, Nano Banana (in the Gemini app), prompt card, class leaderboard	
Procedure	Three rounds of about five minutes each. Teams write their prompts, the trainer runs the best few on the projector and the class votes. • Round 1, Explain It: make Gemini explain photosynthesis so that a Class 3 child understands it, in exactly five lines. • Round 2, Format Master: get a table of five Indian states with their capital, main language and one famous dish, using a single prompt. • Round 3, Recreate the Image: a picture is shown on screen, and teams try to recreate it as closely as possible with Nano Banana.	
Output	Points on the leaderboard, and the R-T-C-F prompt card (Annexure B).	
Concept learnt	Role, Task, Context and Format, and how clear instructions change the result.	
Wow moment	One extra line in a prompt completely changes what the AI produces.	

Mission 4: Build the App		35 minutes
Aim	To build and share a working web app for the team's problem without writing code by hand.	
Tools	Gemini with Canvas; Lovable, shown by the trainer as the advanced option	
Procedure	1. Fill in the app plan on the worksheet: name of the app, who it is for, and three features. 2. Turn on Canvas in Gemini and write the build prompt using the template on the prompt card. 3. Test the app in the preview and note one thing that needs fixing. 4. Improve the app in three rounds: add a feature, fix a problem, improve the design. 5. Add the footer "Made by Team ___ at the BGIEM AI Innovation Lab". 6. Share the app, make a QR code for the link and open it on a phone.	
Output	A working web app, such as a quiz, a planner, a checklist or a calculator, with a shareable link.	
Concept learnt	How AI writes code, why testing matters, and why logic is still needed.	
Stretch challenge	Teams that finish early try to connect their Teachable Machine model to the app to make a "smart bin" page.	
Wow moment	An app built in half an hour, running on a teacher's phone.	

Mission 5: Brand It		20 minutes
Aim	To design the identity and the pitch material for the team's project.	
Tools	Nano Banana (Gemini app), Gemini Canvas	
Procedure	1. Write the 8-second promo video prompt first and hand it to the co-trainer, who starts generating the video in Google Flow. 2. Create a logo for the app and improve it with two follow-up instructions. 3. Create an A4 poster with the app's name, a tagline and space for the QR code. 4. Make a single pitch slide covering the problem, the solution, how it works and the team members.	
Output	Logo, poster, pitch slide and video prompt.	
Concept learnt	Text-to-image generation, editing images with words, and labelling AI-made content.	
Wow moment	A professional-looking poster created in a few minutes.	

Mission 6: Automate It		10 minutes
Aim	To understand automation and AI agents.	
Tools	Team worksheet; Google Workspace Studio (demonstration by the trainer)	
Procedure	1. Each team plans one automation for its project in the form Trigger → Steps → Result. For example: when someone fills our feedback form → save the answers in a Sheet → send a thank-you email. 2. Two teams read out their plans. 3. The trainer builds the best plan live in Google Workspace Studio and runs it in front of the class.	
Output	An automation plan on the worksheet.	
Concept learnt	Triggers, steps, AI agents, and the need for human supervision.	
Wow moment	A teacher fills the form on a phone and the email appears on the screen a moment later.	

Demo Day and Closing		35 minutes
Aim	To present the project with confidence and to close the workshop with a commitment to safe use of AI.	
Format	Each team gets two minutes: the problem, a live demo of the app on the projector, the poster, and the premiere of its AI promo video. The panel consists of the principal or teachers nominated by the school, and the BGIEM trainer.	
Judging	Usefulness of the idea (25), working app (25), use of AI tools (20), design and poster (15), presentation (15). Total: 100 marks.	
Awards	Best AI Innovation, Best App, Best Prompt Engineers (from the Mission 3 leaderboard), Best Design, and Teachers' Choice.	
Closing	All students read the AI Safety Pledge together, receive certificates and the AI Starter Kit, and pose for a group photograph.	
Wow moment	Each team sees its own AI-made promo video on the big screen for the first time.	

5. PROBLEM STATEMENT CARDS

Each team picks one card at the start of the workshop. The school is welcome to replace any card with a problem of its own.

No.	Problem	Possible app idea
1	Waste is not segregated properly on the school campus	Smart bin guide that tells which waste goes into which bin, with a quiz
2	Visitors to Bhedaghat and Dhuandhar Falls do not find information easily	Tourist guide with places to see, safety tips and a short quiz
3	Exam stress and poor study planning	Study planner with a timetable maker and short breathing exercises
4	Pollution in the Narmada and in local water bodies	Water-saving calculator and an awareness quiz
5	Crowding and road safety at the school gate during dispersal	Road safety quiz and a safe-route checklist
6	Students find it hard to locate books in the library	Book finder and recommendation app
7	Healthy eating in the school canteen	Nutrition guide for common canteen items
8	Little awareness of the heritage of Jabalpur, such as Madan Mahal Fort and Rani Durgavati	Heritage quiz and timeline app

Any facts used inside an app, such as timings or nutrition values, are checked by the team from a reliable source before the app is shared.

6. TAKE-HOME AI STARTER KIT

Every team receives a digital folder, shared through a QR code, containing:

- the link to its Teachable Machine model and to its web app;
- its NotebookLM notebook and AI podcast;
- its logo, poster, pitch slide and AI promo video;
- e-certificates for every team member.

Every student also receives a printed R-T-C-F prompt card, the AI Safety Pledge and the 30-Day AI Explorer Challenge:

Week	Challenge
1	Retrain the waste model with three new classes (glass, metal and e-waste) and test it at home.
2	Make a NotebookLM notebook for one school subject and revise one chapter with its quiz and flashcards.
3	Add two new features to the team's app and share it with the class.
4	Teach one family member how to use AI safely, and write a short note on what you learnt.

7. AFTER THE WORKSHOP: CONTINUATION PLAN

7.1 Teacher Handover

A 30-minute session for the school's computer teachers, held on the day of the workshop or a day before. Teachers receive the lab manual (the mission sheets in Section 4), the account setup guide, the problem cards and the worksheets, so that they can repeat the practicals with other classes.

7.2 AI Explorers Club (Four Weeks)

One 60-minute lab session every week, run by the school's computer teacher with a BGIEM mentor:

Week	Theme	Lab work
1	Better data, better models	Sound and pose projects in Teachable Machine; improving model accuracy
2	AI for exams	A NotebookLM study notebook for each subject, with quizzes and flashcards
3	App upgrade	New features, testing with classmates, and publishing the final version
4	Showcase preparation	Final pitch, poster and video for the inter-school showcase

7.3 Inter-school AI Showcase at BGIEM

The best teams from both schools present their projects at the BGIEM campus, visit the Institute's engineering labs and receive awards. This gives students a real audience for their work and a first look at engineering education.

7.4 Mentoring

Follow-up questions are handled through a Google Classroom set up and managed by the school, with BGIEM mentors added as guest teachers.

8. PREPARATION AND LAB REQUIREMENTS

Item	Requirement	Responsibility
Computer lab	One computer per team (10 to 12 computers for 40 students) with an updated Chrome browser	School
Webcams	At least one webcam per team; BGIEM brings six spare USB webcams	School and BGIEM
Internet	Stable connection; 50 Mbps or more is recommended for 40 users	School
Website access	The following sites unblocked in the school firewall: quickdraw.withgoogle.com, teachablemachine.withgoogle.com, gemini.google.com, notebooklm.google.com, studio.workspace.google.com and labs.google	School IT team
Accounts	School Google Workspace for Education accounts with Gemini and NotebookLM turned on by the school's admin; otherwise, team accounts managed by BGIEM	School and BGIEM
Display and sound	Projector, speakers and a microphone in the lab	School
Materials	Waste items for Mission 1, problem cards, source packs, team worksheets, prompt cards, certificates and prizes	BGIEM
Backup	Mobile hotspot, screen recordings of each mission, and sample outputs ready in case a tool is slow	BGIEM

Preparation timeline

- **One week before:** confirm the date, the batch and the number of students with the school; share the website list with the school's IT team.
- **Three to four days before:** visit the lab, test every website and account on two computers, and check the webcams.
- **Two days before:** receive the student list and form balanced teams; print worksheets, cards and certificates.
- **On the day:** reach one hour early, log in on all computers, and test the projector and sound.

9. SAFETY AND RESPONSIBLE USE

- No student creates a personal account during the workshop; only school-managed or BGIEM-managed logins are used.
- No faces or personal data are used to train models. Mission 1 uses only objects.
- Photographs of students are taken only as per the school's permission policy.
- Every AI-made image, poster and video is labelled as AI-generated.
- Facts used in student apps are checked from reliable sources before the apps are shared.
- The workshop ends with the AI Safety Pledge: check before you trust, keep personal information private, use AI to learn and not to copy, respect others, and follow age rules.

10. TWO-HOUR VERSION

If the school can give only two hours of lab time, the workshop runs as follows. The Prompt Battle is dropped and its tips are given during Mission 4.

Time	Activity
0:00–0:05	Opening and team formation
0:05–0:20	Mission 1: Teach a Machine
0:20–0:35	Mission 2: Research Like a Scientist (mind map and podcast)
0:35–1:05	Mission 4: Build the App
1:05–1:20	Mission 5: Brand It (logo and poster)
1:20–1:25	Mission 6: Automation demonstration by the trainer
1:25–1:50	Demo Day: 90-second pitch per team, with promo videos
1:50–2:00	AI Safety Pledge, certificates and group photograph

Mr. Deepak Vishwakarma

Trainer, BGIEM (Prepared by)

Date: 05/10/2026

Dr. Niharika Yadav

Deputy Director, BGIEM (Submitted to)

ANNEXURE A: TEAM MISSION WORKSHEET

One copy per team. Filled in during the workshop and kept by the team.

Team name _____	Problem card number and title _____
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Team members _____

Mission 1: Our model's accuracy before and after the bias test _____ _____

Mission 2: Three important facts about our problem (with source) _____ _____ _____
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Mission 4: Our app plan (name, users and three features) _____ _____ _____
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Mission 5: Our promo video prompt (subject, action, place, camera, light, sound) _____ _____

Mission 6: Our automation (Trigger → Steps → Result) _____ _____

Links: model, app and starter-kit folder _____ _____

ANNEXURE B: PROMPT CARD

Printed for every student. The R-T-C-F method and ready templates for the missions.

Letter	Meaning	Example
R, Role	Who the AI should act as	"You are my science teacher."
T, Task	What exactly you want	"Explain photosynthesis."
C, Context	Level, purpose and details	"I am in Class 8 and have a test tomorrow."
F, Format	How the answer should look	"Five short points and one daily-life example."

App template (Mission 4)

Build a simple, mobile-friendly web app called [app name] for [who it is for]. It should help them [solve which problem]. Include these features: 1) [feature], 2) [feature], 3) [feature]. Use bright colours, large buttons and simple English. Add a footer: "Made by Team [name] at the BGIEM AI Innovation Lab".

Logo template (Mission 5)

Design a simple flat logo for an app called [app name] that helps [purpose]. Use [two colours] on a plain white background, with no extra text.

Poster template (Mission 5)

Create an A4 portrait poster for [app name]. Headline: "[tagline]". Show [a scene related to the problem]. Leave an empty square at the bottom right for a QR code.

Video template (Mission 5)

An 8-second shot of [subject] [doing what] at [place]. The camera [moves how]. [Lighting]. Sound of [sound].

Follow-up prompts that always help

"Make it simpler." "Add one more feature: ..." "Something is not working: ... Please fix it." "Make the design more colourful and the text larger." "Explain this in Hindi."